STATISTICS · STATISTICAL INFERENCE

Confidence Interval Lab

Explore & Reflect

Name _____ Date _____ Period _____

Open **Confidence Interval Lab** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. If you make 100 different 95% confidence intervals from 100 random samples, how many do you expect to capture the true value?

2. Is a 99% interval from the same sample wider or narrower than a 95% interval? Why?

2 Explore use the tool

1. Proportion, p = 0.4, n = 40, 95%: make 100 or more intervals. Record how many you made and how many captured p.

2. Same n: record the average margin of error at 90% and at 99%.

3. At 95%, what n cuts the average margin of error from n = 40 roughly in half?

4. Mean, n = 5: make 200+ intervals with z* and s, then 200+ with t*. Record both capture rates and explain the difference.

3 Record use the tool and fill in the table

Margins of error

Use the formulas shown in the tool's readout.

Setting	Critical value	Standard error	Margin of error
$\hat{p}$ = 0.4, n = 40, 95%			
$\hat{p}$ = 0.4, n = 160, 95%			
$\hat{p}$ = 0.4, n = 40, 99%			
$\bar{x}$, s = 10, n = 5, 95%			
$\bar{x}$, s = 10, n = 30, 95%			

4 Explain what did you discover?

1. "95% confident" means that about 95% of _____ capture the parameter.
2. Margin of error = _____ × _____.
3. To halve the margin of error, use about _____ times the sample size.
4. When σ is unknown, use _____ with df = _____.
5. From one sample, Ana computes the 95% interval (0.31, 0.47) and says, "There is a 95% probability that p is between 0.31 and 0.47." What is wrong? Write a correct interpretation.

5 Apply a new situation, no tool

1. In an SRS of 500 voters, 280 support a measure. Construct a 95% confidence interval for p. (Conditions: 280 and 220 are both ≥ 10.)

2. How large a sample gives a margin of error of at most 0.03 at 95% confidence with no prior estimate of p?

3. An SRS of 12 bottles has $\bar{x}$ = 501.2 mL and s = 2.4 mL. Construct a 95% confidence interval for μ.